

Request for Information (RFI): Priority Policy Actions for the U.S. AI Action Plan

Submitted to: The Office of Science and Technology Policy (OSTP)

Submitted by: NJEdge

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Subject: Recommendations for AI Policy Actions in Research and Development, Education and Workforce, Innovation and Competition, Intellectual Property, and International Collaboration

I. Introduction

The rapid evolution of artificial intelligence (AI) presents both opportunities and challenges for the United States. To maintain global leadership in AI, the U.S. must adopt forward-thinking policies that drive research and development, strengthen education and workforce pipelines, promote innovation and fair competition, protect intellectual property, and foster international collaboration. This Request for Information (RFI) response outlines key policy actions needed in these areas to ensure the responsible and sustainable advancement of AI technologies.

II. Research and Development (R&D)

A. Increased Federal Investment in AI R&D

- Expand funding for fundamental AI research through agencies such as NSF, DOE, DARPA, and NIH.
- Establish dedicated R&D programs focused on ethical AI, explainability, and robustness of AI systems.
- Support AI research in high-performance computing, edge computing, and neuromorphic computing.
- Increase investment in domain-specific AI applications in sectors where the U.S. has strong capabilities, including healthcare, cybersecurity, energy, aerospace, and defense.
- Promote AI research in autonomous systems, including robotics, self-driving vehicles, and AI-driven industrial automation.
- Fund interdisciplinary research initiatives that merge AI with quantum computing, biotechnology, and materials science to drive breakthrough innovations.

B. Public-Private Partnerships

- Strengthen collaboration between government, industry, and academia to accelerate AI breakthroughs.
- Develop AI innovation hubs that provide shared resources and infrastructure for AI research.
- Incentivize industry participation in AI research consortia.
- Establish federally supported AI testbeds to enable real-world AI experimentation in controlled environments.

C. Infrastructure Development

- Expand access to national AI research cloud platforms and high-performance computing resources.
- Support the development of AI-specific data-sharing repositories with standardized protocols.
- Invest in energy-efficient AI computing infrastructure to mitigate environmental impact.
- Enhance AI-driven cybersecurity measures for protecting AI models and datasets.
- Develop AI-powered simulation and digital twin technologies to enhance industrial and scientific R&D.

D. Workforce Development for AI Research

- Establish targeted funding for AI workforce development programs to build a skilled talent pool in AI-related disciplines.
- Create federally funded AI training programs at universities, community colleges, and vocational schools to develop AI expertise at various skill levels.
- Develop AI research fellowships and scholarships for underrepresented groups to promote diversity in AI research.
- Support industry-academic partnerships to create AI apprenticeship and internship programs.
- Establish training centers to upskill government employees in AI applications to enhance public sector AI readiness.
- Encourage AI-related continuing education programs for professionals to stay current with emerging AI trends and technologies.

III. Education and Workforce

A. AI Education in K-12 and Higher Education

- Integrate AI literacy into K-12 curricula to prepare students for future AI-driven careers.
- Increase federal funding for AI-related higher education programs and interdisciplinary AI research.

- Promote AI training and reskilling programs through community colleges and technical institutions.

B. Workforce Development and Training

- Expand apprenticeships, internships, and fellowships in AI across public and private sectors.
- Support continuous learning initiatives for AI professionals through federal workforce grants.
- Address AI-related job displacement by funding retraining programs for affected workers.

C. Diversity, Equity, and Inclusion (DEI) in AI Careers

- Implement policies to ensure AI workforce diversity and equitable access to AI education.
 - Provide grants for minority-serving institutions to develop AI-focused programs.
 - Support initiatives that address bias in AI recruitment and hiring processes.
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IV. Innovation and Competition

A. Support for AI Startups and Small Businesses

- Increase funding for AI startups through Small Business Innovation Research (SBIR) and Small Business Technology Transfer (STTR) programs.
- Create AI-specific tax incentives to encourage entrepreneurship and innovation.
- Establish regulatory sandboxes to facilitate responsible AI experimentation and deployment.

B. Fair Competition and Market Regulation

- Implement antitrust measures to prevent AI monopolization and promote healthy competition.
- Ensure transparency in AI procurement processes to support small and mid-sized enterprises (SMEs).
- Establish AI certification programs to promote responsible AI use and consumer trust.

C. AI in Critical Sectors

- Encourage AI-driven innovations in healthcare, finance, education, and manufacturing.
- Develop AI policies that support ethical AI deployment in critical sectors.

- Promote responsible AI adoption in government services and public administration.
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V. Intellectual Property (IP)

A. Modernizing AI Patent and Copyright Policies

- Establish AI-specific patent frameworks that recognize AI-assisted innovation.
- Clarify legal protections for AI-generated content and ensure fair compensation for creators.
- Address challenges in AI data ownership and licensing agreements.

B. Balancing Open Source and Proprietary AI Development

- Develop guidelines for responsible open-source AI development while protecting IP rights.
- Encourage AI model transparency while safeguarding trade secrets and security concerns.
- Establish public-private dialogues to refine AI-related IP policies.

C. Addressing AI's Impact on Traditional IP Laws

- Assess how AI affects existing copyright, trademark, and patent laws.
 - Develop policies to protect AI-generated music, art, and literature.
 - Establish legal frameworks to determine liability in cases of AI-generated IP disputes.
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VI. International Collaboration

A. Strengthening Global AI Partnerships

- Expand AI research partnerships with allied nations and international organizations.
- Promote joint AI research projects and cross-border AI training programs.
- Support U.S. leadership in international AI governance discussions.

B. Establishing AI Standards and Ethical Guidelines

- Collaborate with global stakeholders to create interoperable AI safety and ethics standards.
- Promote responsible AI deployment through multilateral agreements and regulatory harmonization.

- Encourage knowledge-sharing initiatives for AI best practices and risk mitigation.

C. Export Controls and National Security

- Update AI export control policies to prevent misuse while ensuring market competitiveness.
- Strengthen AI-related cybersecurity measures to protect national security.
- Implement policies that prevent adversarial nations from exploiting U.S. AI advancements.

VII. Conclusion and Policy Recommendations

The U.S. must act swiftly and strategically to maintain leadership in AI while ensuring its ethical and equitable development. We recommend:

1. Expanding federal AI R&D investments and infrastructure.
2. Strengthening AI education, training, and workforce diversity.
3. Encouraging fair competition, innovation, and regulatory clarity.
4. Modernizing intellectual property frameworks for AI-generated content.
5. Enhancing international AI collaborations and standardization efforts.

By adopting these policies, the U.S. will ensure AI's responsible growth, benefiting society while maintaining its competitive edge. We appreciate the opportunity to contribute to the OSTP's AI Action Plan and look forward to future discussions.

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